

Form Management Workflow Documentation

Health Data Bank System

1. Purpose

All form submission, review, and status transitions in this project must follow the standardized Form Management Workflow.

This ensures:

- Consistent validation handling
- Proper role-based access control (RBAC)
- Controlled approval lifecycle
- Audit logging compatibility
- Privacy and data protection compliance

Forms must never bypass the defined workflow states or middleware protections.

2. Scope

This document applies to:

- Health Data Submission forms
- Administrative approval forms
- Revision/resubmission flows
- Form filtering and management interfaces

3. Core Workflow Model

Every managed form in the system must follow the standardized status lifecycle.

3.1 Status States

All form submissions must use one of the following statuses:

- Draft
- Pending
- Approved
- Rejected
- Returned

Additional statuses must be approved before implementation.

3.2 Status Transition Rules

The following transitions are permitted:

From	To	Allowed By
Draft	Pending	User / Researcher / Provider
Pending	Approved	Administrator
Pending	Rejected	Administrator
Pending	Returned	Administrator
Returned	Pending	Original Submitter

Transitions not listed above are prohibited.

4. Role-Based Form Permissions

Form access must be enforced using middleware and policy checks.

4.1 User / Researcher / Provider

Allowed:

- Create new submission
- Edit Draft
- Edit Returned submissions
- View own submissions

Not Allowed:

- Modify Approved submissions
- Approve or Reject submissions
- Access other users' submissions

4.2 Administrator

Allowed:

- View all submissions
- Approve submissions
- Reject submissions
- Return submissions for revision
- Filter and search submissions

Not Allowed:

- Modify user-submitted content directly
- Edit approved historical records

5. Submission Processing Rules

5.1 On Initial Submission

When a form is submitted:

1. Validate request input
2. Store submission in database
3. Set status = Pending
4. Create audit log entry
5. Trigger notification (if enabled)
6. Redirect to dashboard

Submission must not proceed if validation fails.

5.2 On Validation Failure

The system must:

- Prevent database write
- Display inline field errors
- Preserve previously entered input
- Display flash error message (if applicable)

6. Edit & Revision Workflow

6.1 Editable Conditions

A submission may only be edited if:

- Status = Draft

OR

- Status = Returned

Editing is strictly prohibited when:

- Status = Pending
- Status = Approved

6.2 Resubmission

When a Returned form is edited and resubmitted:

1. Status resets to Pending
2. Previous version is preserved (if versioning enabled)
3. Audit entry recorded
4. Admin review queue updated

7. Administrative Review Process

7.1 Review Interface Requirements

Admin panel must provide:

- Read-only view of submission data

- Decision selector (Approve / Reject / Return)
- Comment field (required for rejection or return)

7.2 On Approval

System must:

1. Update status to Approved
2. Record audit entry
3. Notify submitter
4. Lock submission from further edits

7.3 On Rejection

System must:

1. Update status to Rejected
2. Record rejection reason
3. Log audit entry
4. Notify submitter

Rejected submissions remain immutable.

7.4 On Return for Revision

System must:

1. Update status to Returned
2. Store admin comments
3. Log audit entry
4. Notify submitter

Returned submissions become editable.

8. Form Validation Standards

All forms must enforce:

- Required field validation
- Format validation (email, numeric, date)
- Length restrictions
- File type restrictions (PDF/JPG/PNG)
- File size limits

Validation rules must be defined in:

- Controller validation logic

OR

- Form Request classes

Client-side validation may enhance UX but does not replace server-side validation.

9. Data Integrity & Security Rules

All forms must:

- Use CSRF protection
- Enforce RBAC middleware
- Prevent ID tampering
- Restrict mass assignment
- Sanitize file uploads

Sensitive data must not be exposed in error messages.

10. Notifications

The system may trigger:

- Email notifications
- In-app dashboard alerts

Notifications must be triggered only after successful state changes.

11. Database Requirements

Submissions must be stored using relational integrity.

Required tables:

- users

- roles
- health_forms (or submissions)
- form_versions (if versioning enabled)
- audit_logs
- approval_logs

Foreign keys must be enforced.

12. When to Apply This Workflow

This workflow must be followed when:

- A new form type is introduced
- Approval logic is added
- A submission status is modified
- Role-based restrictions are implemented
- Admin moderation features are developed

Direct database writes that bypass status lifecycle logic are prohibited.

13. Compliance & Governance

The Form Management Workflow supports:

- Accountability
- Data integrity
- Privacy-by-design
- Audit compatibility
- Secure approval processes

All future enhancements must remain compatible with this workflow structure.

14. Actual Form Workflow in the System

This section describes the real workflow followed when a user submits and manages a form in the Health Data Bank system.

Step 1 – Form Creation

A user (User, Researcher, or Healthcare Provider) creates a new form submission through the UI.

The system validates the input fields and stores the form in the database with status Draft.

Step 2 – Form Submission

When the user submits the form:

1. The system validates all required fields.
2. If validation succeeds, the form status changes to Pending.
3. The submission is stored in the database.
4. An audit log entry is recorded.

Step 3 – Admin Review

Administrators review submitted forms from the admin dashboard.

The administrator can choose one of three actions:

- Approve
- Reject
- Return for revision

The decision is recorded in the system and logged for audit purposes.

Step 4 – Approval

If the administrator approves the submission:

- Status changes to Approved
- The form becomes read-only
- An audit log entry is created
- The submitter is notified

Approved forms cannot be edited.

Step 5 – Rejection

If the administrator rejects the submission:

- Status changes to Rejected
- The rejection reason is recorded
- The user is notified
- The form cannot be modified

Step 6 – Return for Revision

If the administrator returns the form:

- Status changes to Returned
- Admin comments are stored
- The submitter can edit the form again

When the form is edited and resubmitted, the status returns to Pending and the review process repeats.

Step 7 – Audit Logging

All workflow actions generate audit log entries, including:

- form submission
- status changes
- admin decisions
- resubmissions

This ensures traceability and compliance with system governance requirements.